

FRITZLAR, Germany

WMO: 104390

Lat: 51.1189N

Lon: 9.2800E

Elev: 181

StdP: 99.16

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.0	-13.0	1.3	-9.3	-10.4	1.6	-7.1	10.7	6.4	9.4	6.7	1.7	310	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.5	20.0	28.2	19.0	26.2	18.1	20.9	28.1	19.9	26.5	19.0	24.9	3.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.6	23.2	17.7	13.0	22.5	16.8	12.3	21.4	61.0	28.0	57.7	26.4	54.7	24.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.3	6.3	-13.4	33.7	3.7	1.8	-16.1	35.0	-18.3	36.1	-20.4	37.1	-23.1	38.5
			-13.7	22.8	3.6	1.1	-16.3	23.6	-18.4	24.2	-20.4	24.8	-23.0	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	1.0	2.0	5.3	9.3	13.1	16.3	18.4	18.1	14.2	9.8	5.3	1.9
	DBStd	7.20	4.71	4.32	3.67	3.80	3.55	3.29	3.24	3.10	3.01	3.52	3.55	4.29
	HDD10.0	1171	280	224	151	59	11	1	0	0	3	47	145	251
	HDD18.3	3291	538	457	405	273	165	79	41	42	129	264	391	508
	CDD10.0	1026	0	1	4	37	108	188	261	253	128	42	4	1
	CDD18.3	105	0	0	0	0	4	17	43	36	4	0	0	0
	CDH23.3	1262	0	0	0	9	66	219	487	404	74	2	0	0
	CDH26.7	381	0	0	0	1	9	54	166	136	15	0	0	0
Wind	WSAvg	2.7	3.1	3.0	3.1	2.8	2.6	2.6	2.5	2.4	2.4	2.5	2.7	3.0
Precipitation	PrecAvg	670	52	43	43	43	64	66	75	62	55	53	58	55
	PrecMax	893	108	108	79	80	151	115	131	137	124	153	88	120
	PrecMin	489	4	12	6	3	18	28	22	19	6	9	2	15
	PrecStd	95	26	24	21	18	34	21	29	30	28	30	23	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.1	18.6	24.7	27.9	31.0	33.3	33.3	28.9	21.9	15.2	12.6
		MCWB	9.7	8.9	11.2	14.8	18.5	20.6	20.4	20.1	18.9	16.5	12.4	10.1
	2%	DB	10.1	11.1	15.8	21.8	25.1	28.1	30.9	30.2	25.7	18.7	13.5	10.9
		MCWB	8.1	8.2	10.0	13.8	16.9	19.7	20.0	19.5	17.5	15.0	11.2	9.1
	5%	DB	8.8	9.8	13.2	19.2	22.9	25.9	28.1	27.7	22.6	16.8	12.0	9.2
		MCWB	7.2	7.4	9.0	12.4	15.3	18.3	19.1	18.6	16.4	14.0	10.1	7.8
	10%	DB	7.2	8.2	11.2	16.9	20.8	23.8	26.0	25.1	20.1	15.2	10.5	7.9
		MCWB	5.8	6.2	8.1	11.4	14.5	17.2	18.2	17.7	15.2	12.9	8.9	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	9.8	12.7	15.9	20.1	22.2	22.1	21.4	19.7	16.8	13.1	10.7
		MCDB	11.7	11.7	16.5	22.0	26.4	28.6	29.5	29.6	26.2	20.4	14.5	12.1
	2%	WB	8.4	8.7	10.8	14.4	17.9	20.4	20.9	20.4	18.2	15.5	11.5	9.3
		MCDB	9.8	10.8	14.2	20.4	23.4	26.4	28.1	27.9	23.9	18.2	13.3	10.6
	5%	WB	7.1	7.5	9.6	13.0	16.4	19.1	19.9	19.4	17.0	14.3	10.2	7.8
		MCDB	8.5	9.5	12.9	18.1	21.5	24.7	26.5	25.9	21.6	16.4	11.7	9.2
	10%	WB	5.7	6.3	8.3	11.8	15.0	17.8	18.9	18.4	15.9	13.2	9.1	6.6
		MCDB	7.1	8.2	11.2	16.2	19.3	22.7	24.7	23.8	19.5	15.2	10.4	7.7
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.7	8.6	10.8	11.0	11.2	11.4	11.6	10.2	8.2	5.5	4.8
		MCDBR	5.4	8.9	13.1	15.4	15.4	15.6	16.1	16.4	15.0	10.5	7.0	5.6
		MCWBR	4.5	6.3	8.2	8.4	8.0	7.9	6.8	7.0	7.8	6.9	5.4	4.7
	5% WB	MCDBR	5.3	7.8	11.6	13.5	13.7	14.2	14.1	13.8	12.6	9.8	6.6	5.5
		MCWBR	4.7	5.9	7.6	7.6	7.7	7.6	6.6	6.5	6.9	6.8	5.3	4.8
Clear-Sky Solar Irradiance	taub	0.309	0.333	0.368	0.406	0.411	0.409	0.422	0.415	0.382	0.354	0.333	0.306	
	taud	2.411	2.368	2.300	2.229	2.250	2.272	2.241	2.282	2.360	2.425	2.421	2.406	
	Ebn at Noon	716	791	825	831	845	847	829	813	802	756	675	655	
	Edh at Noon	61	82	105	126	130	128	130	119	99	77	60	53	
All-Sky Solar Radiation	RadAvg	0.72	1.40	2.51	4.01	4.79	5.25	4.98	4.27	3.07	1.79	0.81	0.53	
	RadStd	0.10	0.22	0.35	0.50	0.52	0.49	0.65	0.40	0.42	0.19	0.12	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	-119	+48	N/A

Nomenclature: See separate page